

4.13.86

AI25BTECH11017-BALU

Question:

A plane which is perpendicular to two planes

$$2x - 2y + z = 0 \quad \text{and} \quad x - y + 2z = 4 \quad (0.1)$$

passes through $(1, -2, 1)$. The distance of the plane from the point $(1, 2, 2)$ is (2006).

Solution:

Let us solve the given equation theoretically and then verify the solution computationally

According to the question,

Given two planes

$$\mathbf{n}_1 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \quad \mathbf{n}_2 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \quad \mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \quad (0.2)$$

Let $\mathbf{n}$ be normal vector required plane

$$\mathbf{n}_1^T \mathbf{n} = 0 \quad (0.3)$$

$$\mathbf{n}_2^T \mathbf{n} = 0 \quad (0.4)$$

$$\begin{pmatrix} \mathbf{n}_1 & \mathbf{n}_2 \end{pmatrix}^T \mathbf{n} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.5)$$

From solving we

$$\mathbf{n} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad (0.6)$$

Equation of plane is

$$\mathbf{n}^T \mathbf{x} = c \quad (0.7)$$

$$\mathbf{n}^T \mathbf{a} = c = -1 \quad (0.8)$$

distance of point from plane is given by

$$k = \frac{|\mathbf{n}^T \mathbf{x} - c|}{\|\mathbf{n}\|} \quad (0.9)$$

$$k = \frac{|\mathbf{n}^T \mathbf{b} - c|}{\|\mathbf{n}\|} = \frac{4}{\sqrt{2}} \quad (0.10)$$

Highlighted Plane with Point-to-Plane Distance

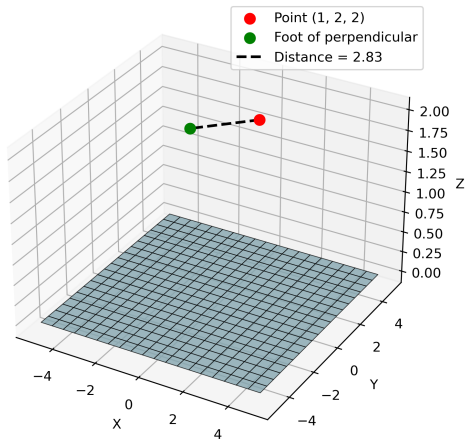


Fig. 0.1